PHILIPS

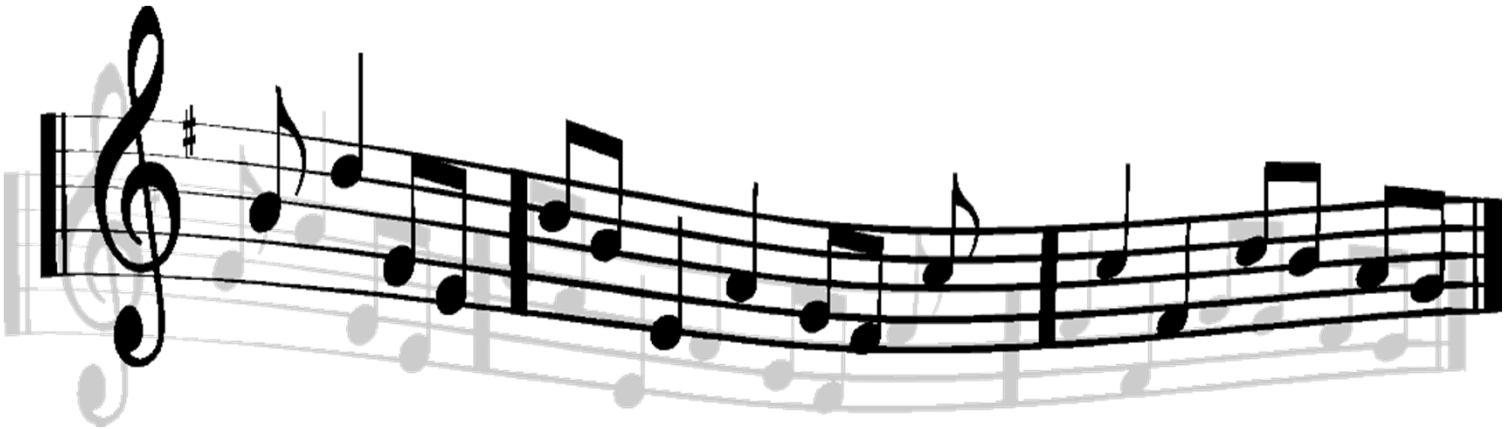
DICOM Data Encoding

Agenda

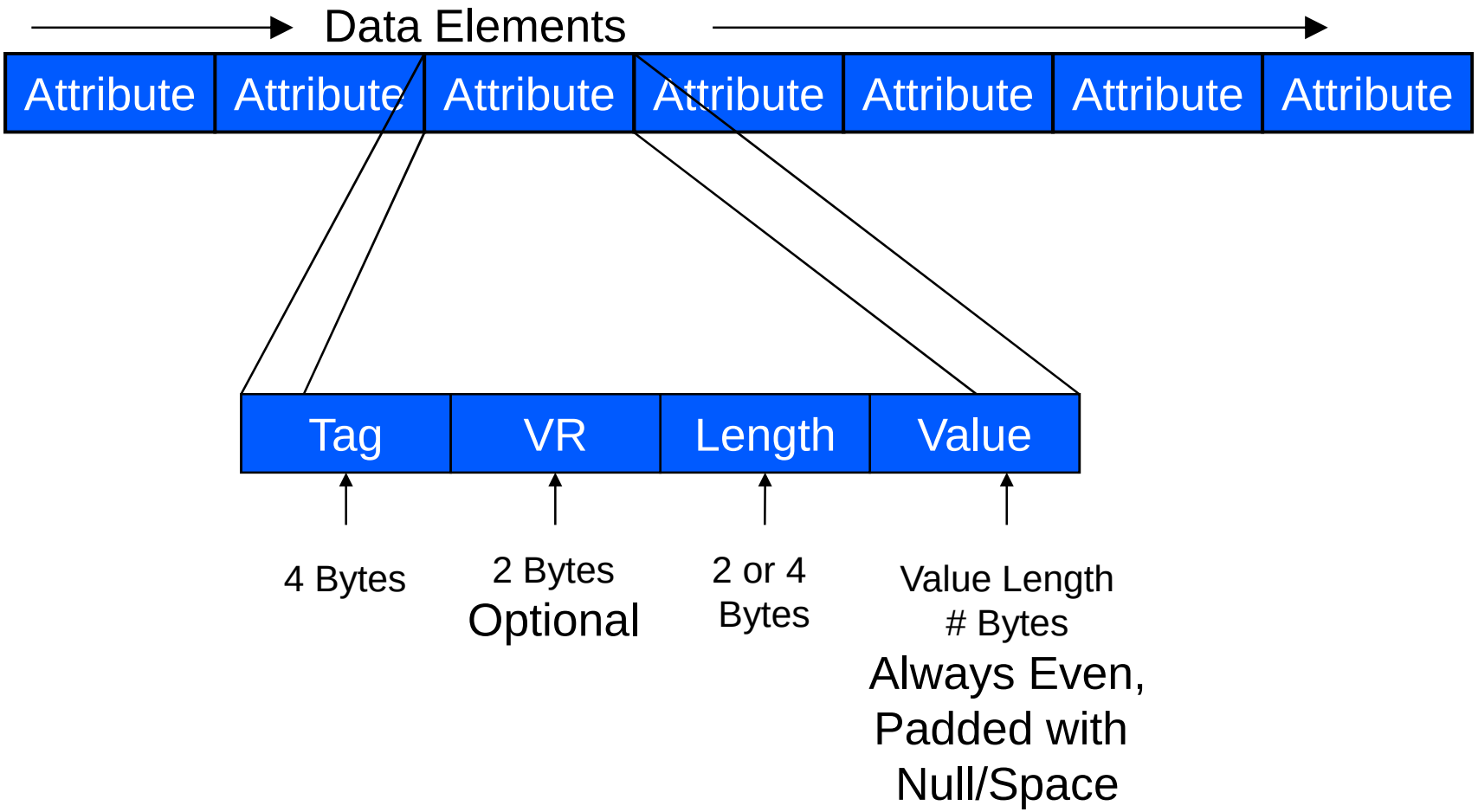
- Data Set
- Data Encoding
- Transfer Syntax
- Unique Identifiers
- Demo

Data Set

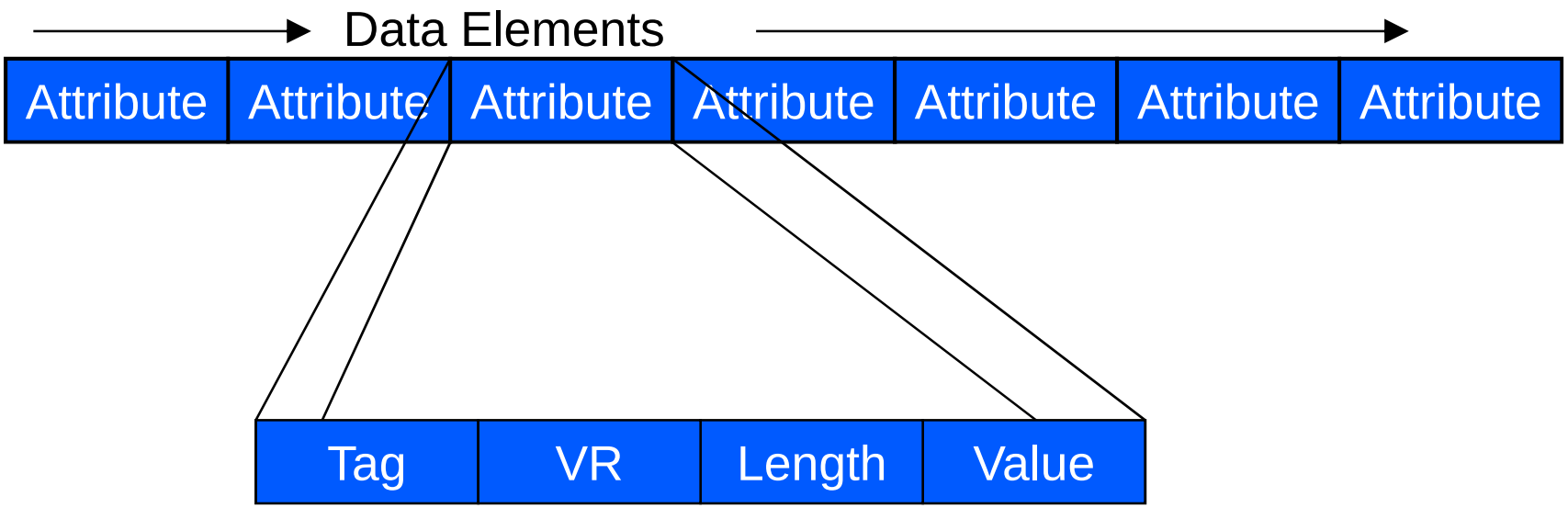
- Structural representation of Data
  - e.g: IOD
- Sequence of Attributes
- Encoded in ascending order of Tags



Data Set Encoding



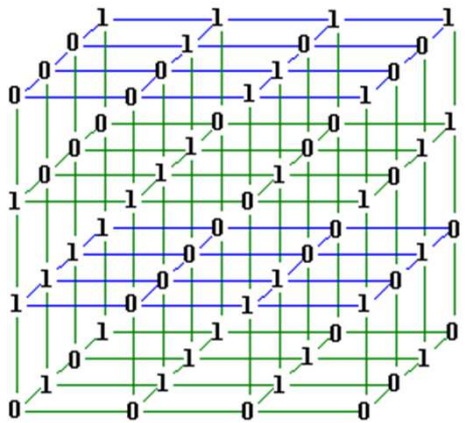
Explicit Encoding



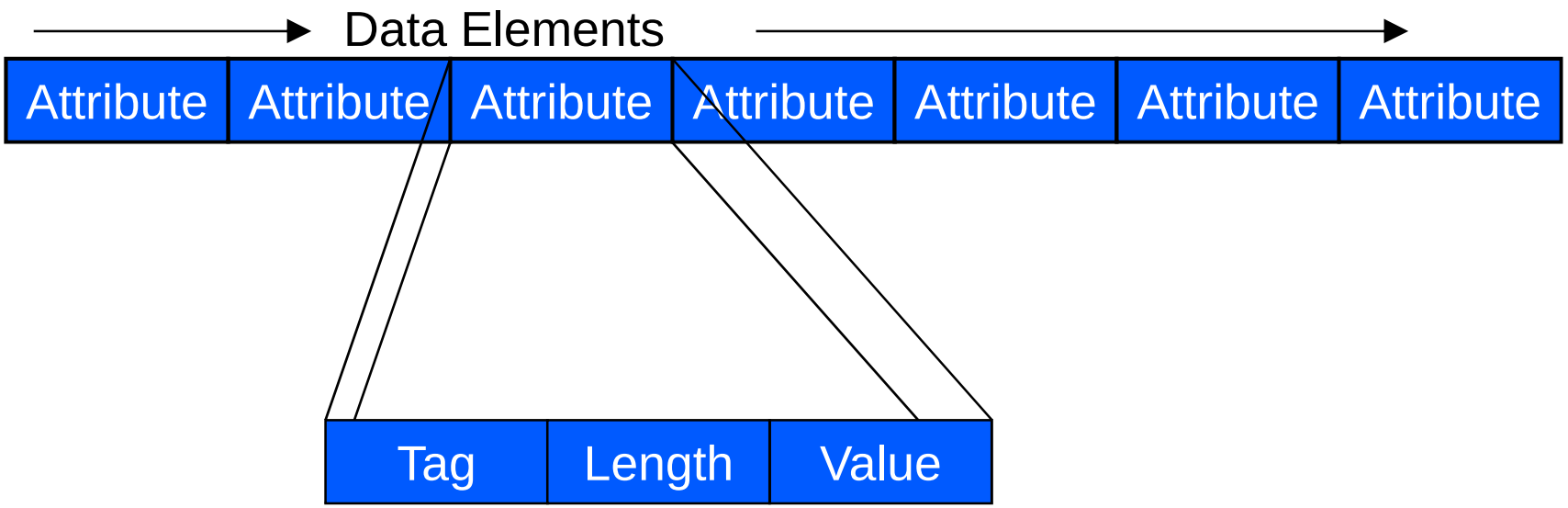
Explicit VR

Explicit Encoding

- Explicit Value Representation
  - VR is a Part of the Data Set
  - Value Length is 4 Bytes for OB/OW/OF/SQ/UT/UN
  - No need to maintain a Dictionary
  - Very Helpful for dealing with Private Data



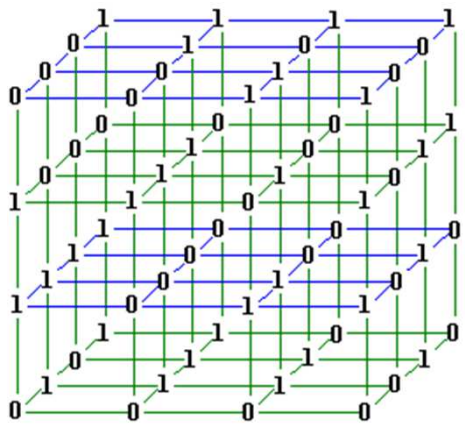
Implicit Encoding



Implicit VR

Implicit Encoding

- Implicit Value Representation
  - VR is not a Part of the Data Set
  - Value Length is always 4 Bytes
  - VR’s are deduced based on Data Dictionary



Byte Ordering

- WORDS in Computer Memory
  - 2 Bytes Unsigned Short Integer
  - 4 Bytes Unsigned Integer
  - 4 Bytes Single Precision Binary Floating Point Number
  - 8 Bytes Unsigned Long Integer
- What are different ways of storing WORDS?
  - Intel Representation
  - Motorola Representation

Intel way of Encoding

- Also known as Little Endian
  - Computer Words are stored such that Least Significant Bytes comes first
  - Example
    - Decimal 12345678
    - Hex BC614E
    - Little Endian 4E 61 BC 00

عيشي بلادي عاش اتحاد إماراتنا
عشت لشعب دينه الإسلام هديه القرآن
حصنتك بسم الله يا وطن
دام الأمان وعاش العلم يا إماراتنا
رمز العروبة كلنا نفديكِ
بالدما نرويكِ
نفديك بالأرواح يا وطن.

Motorola way of Encoding

- Also known as Big Endian
  - Computer Words are stored such that Most Significant Bytes comes first
  - Example
    - Decimal 12345678
    - Hex BC614E
    - Big Endian 00 BC 61 4E

हालांकि सूर के जीवन के बारे में कई जनश्रुतियाँ प्रचलित हैं, पर इन में कितनी सच्चाई है यह कहना कठिन है। कहा जाता है उनका जन्म सन् १४७८ में दिल्ली के पास एक ग़रीब ब्राह्मीण परिवार में हुआ। जनश्रुति के अनुसार सूरदास जन्म से ही अंधे थे। आजकल थी अंधे आदमी अक्सर 'सूरदास' कहलाते हैं। कई लोगों ने उन्हें गुरु के रूप में अपनाया और उनकी पूजा करना शुरु कर दिया ।

Byte Ordering

- Words Transferred from Big Endian Computer to Little Endian Computer require Byte Swapping

- Strings are independent of Endianship

  e.g: AE, AS, CS, DS, PN, SH, ST etc…

Transfer Syntax

केयूराद्यैर्भूषितो वा ततः किं कौशेयाद्यैरावृतो वा ततः किम्।
तृप्तो मृष्टान्नादिना वा ततः किं येन स्वात्मा नैव साक्षात्कृतोऽभूत्।।२।।

2) The body has been adorned with bracelets (and other ornaments), it has been clad in silken (and other rich) robes, it has been pampered with dainty food - all these, verily are in vain to him, by whom the Self has not been realized.

2) எவனொருவனால் தன்னுடைய ஆத்ம ஸ்வரூபம் சாக்ஷாத்கரிக்கப்படவில்லையோ, அவன் தோள்வளை முதலானவைகளால் அலங்காரம் செய்யப்பட்டு விட்டான், அதனால் என்ன பயன்? பட்டு முதலானவைகளால் போர்த்தப்பட்டு விட்டான், அதனால் என்ன பயன்? சுத்தமான ஆஹாரத்தால் திருப்தி அடைந்து விட்டான், அதனால் என்ன பயன்?

Transfer Syntax

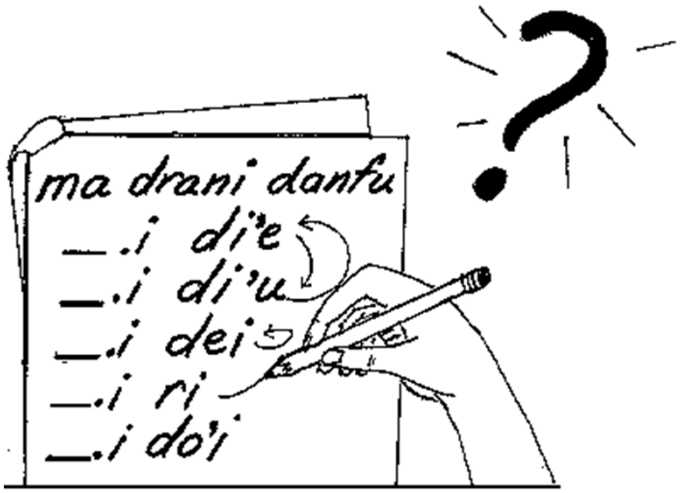
यदा यदा हि धर्मस्य ग्लानिर्भवति भारत ।
अभ्युत्थानमधर्मस्य तदात्मानं सृजाम्यहम् ॥ ७ ॥

Whenever and wherever a decline of righteousness and a predominance of unrighteousness prevails; at that time I manifest Myself personally, O descendant of Bhārata.

ಭರತ ವಂಶಜನಾದ ಅರ್ಜುನನೆ, ಯಾವಾಗ ಎಲ್ಲೆಲ್ಲಿ ಧರ್ಮದ
ಅವನತಿಯಾಗುತ್ತದೆಯೋ ಮತ್ತು ಅಧರ್ಮವು ಹೆಚ್ಚುತ್ತದೆಯೋ ಆಗ ಸ್ವಯಂ
ಅವತಾರ ಮಾಡುತ್ತೇನೆ

Transfer Syntax

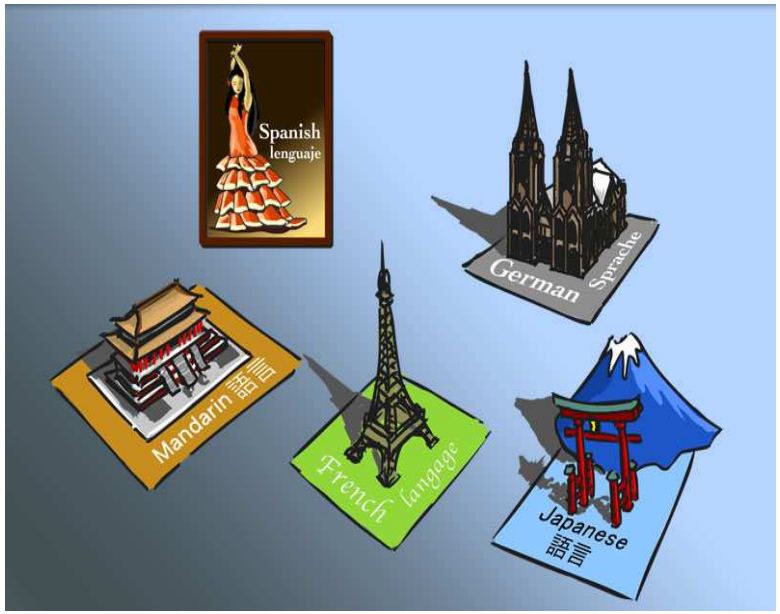
- The Format of Data Encoding
- Transfer Syntax Contains
  - *BYTE ORDERING*
    - Little Endian
    - Big Endian
  - *VALUE REPRESENTATION*
    - Implicit VR
    - Explicit VR
  - *COMPRESSION*
    - JPEG
    - RLE



Transfer Syntax

Few Transfer Syntaxes supported in DICOM

- Implicit VR Little Endian ← DICOM Default Transfer Syntax
- Explicit VR Little Endian
- Explicit VR Big Endian
- JPEG Lossless
- JPEG 2000
- RLE Lossless



DICOM Data Encoding

- Demo
  - Hexadecimal Encoding of an IOD
  - Examples of
    - Value Representation
    - Value Multiplicity

Data Set Encoding: Person Name

- Value Representation PN
- Consists of 5 Components in the Order
  - Last Name, First Name, Middle Name, Prefix, Suffix
  - ^ is the delimiter between each component
  - Any of the components may be an empty string
  - Delimiters are required for null Components
  - Trailing null components and delimiters may be omitted
  - 64 characters max per component group
  - \ character is not allowed

Data Set Encoding: Patient Name

Example

Patient Name

Sri Laloo Prasad Yadav MP Ex CM

Tag

0010 0010

VR

PN

Data Set Encoding Example

– Implicit VR Little Endian

10 00 10 00 32 00 00 00 Yadav^Laloo^Prasad^Sri^MP EX CM

Tag Length Value

– Explicit VR Big Endian

00 10 00 10 **PN** 00 32 Yadav^Laloo^Prasad^Sri^MP EX CM

Tag VR Length Value

– Explicit VR Little Endian

10 00 10 00 **PN** 32 00 Yadav^Laloo^Prasad^Sri^MP EX CM

Tag VR Length Value

Exercise

- Encode your Names in the Following Transfer Syntax
  - Implicit VR Little Endian
  - Explicit VR Big Endian
  - Explicit VR Little Endian
  - JPEG 2000

Private Attributes

- Same structure as DICOM Attributes
- Have Odd group number
- The First Private Attribute is called Private Creator
- Private Creator
  - Reserves a block of tags
  - Has Vendor Specific Recognition Codes
- Private Creator (2001, 0010) Reserves

  2001, 1000 to 2001, 10FF

Private Attributes: Example

Tag	VR	Value
2001,0010	LO	Philips Imaging DD 001
2001,102d	SS	1
2001,1032	FL	0
2001,1033	CS	AP
2001,1036	CS	PARALLEL
2003,0010	LO	Philips EV Imaging DD 001
2003,1071	FL	-13.6
2003,1072	FL	0
2003,1073	FL	16.8
2005,0013	LO	Philips MR Imaging DD 004
2005,1374	FL	278.409
2005,1375	FL	10

Unique Identifiers

- IODs, Transfer Syntax are identified by Unique Identifiers (UID)
  - Identifier of an IOD is SOP Class UID
  - Identifier of Transfer Syntax is Transfer Syntax UID
- Universally Unique
- Assigned by NEMA

Unique Identifiers

UID Format

1.2.840.10008.XXXX

Where

1	ISO Organization
1.2	ISO Member Body
1.2.840	USA Branch of Member Body
1.2.840.10008	DICOM Standard

Similarly

1.2.840.10003	ANSI Standard Z.39
1.2.840.10006	IEEE 802.3

DICOM℠
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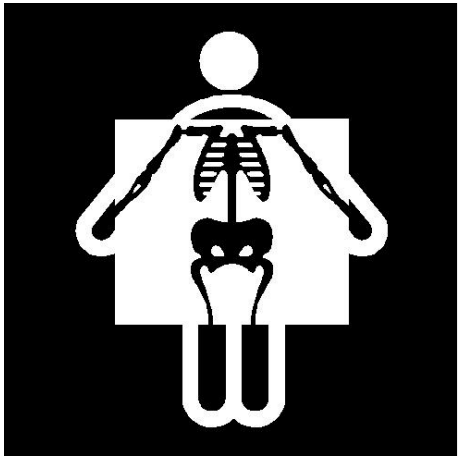
Unique Identifiers

- Image Identifiers are known as SOP Instance UID
- SOP Instance UID
  - Globally Unique
  - NEMA assigns prefix to Vendors
  - Suffix are added by Vendors for Various Products
  - Philips Prefix

    1.3.46.670589



DICOM Data Encoding

- Demo
  - Dump of DICOM Data
  - Implicit VR/Explicit VR
  - Patient Name/Patient ID
  - UIDs

Exercise

- Encode the following as private attributes
  - Your Name, Date of Birth, Sex, Employee ID, Designation, Company Name, LOB Name

  Where Private Creator Tag is 5001 0010 and use your own tags for the attributes mentioned above

- Identify the following information from the Hexadecimal Dump
  - Patient Name & Length 0010, 0010
  - Patient Birth Date, Length and VR 0010, 0030
  - Patient Sex, Length and VR 0010, 0030
  - Identify the Private Creator Attributes

PHILIPS

sense and simplicity